

M Tech (AIML)_Machine Learning _Lab Assignment

Problem Statement

A digital service provider is experiencing gradual customer attrition across its subscription base. While historical customer interaction and service usage data are available, the organization seeks to design a **data-driven system** that can both **predict attrition risk** and support **strategic intervention planning**.

The dataset provided contains anonymized customer-level attributes capturing demographic characteristics, subscription behavior, and service configurations. The data reflects real-world imperfections such as missing values and heterogeneous feature types.

The objective is to develop analytical and machine learning approaches that go beyond naive prediction and provide **actionable insights, robust modeling, and decision-oriented evaluation**.

Dataset Description

The dataset consists of structured customer records with features representing:

- Demographic attributes
- Subscription duration and engagement
- Service configurations and add-ons
- Target variable indicating customer attrition

Q1: Statistical Profiling & Hypothesis Validation

Perform a comprehensive statistical analysis of the dataset.

- Identify key distributional characteristics of features
- Analyze relationships between variables and attrition
- Formulate at least two testable hypotheses regarding drivers of attrition
- Validate hypotheses using appropriate statistical tests
- Critically discuss limitations of inferring causality from observed data

Q2: Data Quality Engineering & Feature Construction

Design a robust preprocessing and feature engineering pipeline.

- Analyze missing data patterns and justify handling strategies
- Detect and address outliers using statistically sound methods
- Apply and compare at least two encoding techniques for categorical variables
- Construct new features that capture latent behavioral patterns
- Perform feature selection using multiple approaches and justify choices

Q3: Predictive Modeling & Comparative Evaluation

Develop and evaluate multiple machine learning models.

- Train at least three classification models with different underlying assumptions
- Perform hyperparameter tuning using cross-validation
- Evaluate models using appropriate performance metrics
- Analyze trade-offs between models in terms of bias, variance, and interpretability
- Justify the final model selection

Q4: Cost-Aware Decision Framework

Reformulate the prediction problem into a decision-making framework.

- Define a cost structure for different types of prediction errors
- Derive an optimal decision threshold based on cost considerations
- Analyze how model performance changes under different threshold choices
- Demonstrate why standard accuracy metrics may be misleading in this context

Q5: Interpretability & Strategic Targeting

Extract actionable insights and design intervention strategies.

- Identify key factors influencing attrition using interpretability techniques
- Rank customers based on predicted risk
- Propose a strategy for prioritizing customers for intervention under limited resources
- Estimate the expected impact of the proposed strategy
- Discuss potential ethical considerations in deploying such a system

Submission Requirements

- Jupyter Notebook containing:
 - Implementation
 - Explanations and justifications
- Concise report (maximum 5 pages) including:
 - Methodology
 - Key findings
 - Insights and conclusions